

Probability Distributions.

Introduction:

In our normal conversations we make statements like 'I am likely to be out of station', 'I have a good chance of being selected to the job' etc. In such statements we are not sure of the outcome but we intend to estimate the chances of our statements being true.

Probability: gives an insight for such things in a mathematical way.

Definitions

Exhaustive event: An event consisting of all the various possibilities is called an exhaustive event.

Mutually exclusive events: Two (or) more events are said to be mutually exclusive if the happening of one event prevent the simultaneous happening of the others.

Examples:

1. In tossing a coin, getting head & tail are mutually exclusive in view of the fact that if head is the turn out, getting tail is not possible.
2. In throwing a cubical 'die', getting any of the no 1, 2, 3, 4, 5, 6 are mutually exclusive as the turn out of any no rules out the possibility

of the turn out of other nos.

Independent events: Two (or) more events are said to be independent if the happening (or) nonhappening of one event does not prevent the happening (or) non happening of the others.

Examples:

1. When two coins are tossed the event of getting head is an independent event as both the coins can turn out heads.
2. When a card is drawn at random from a pack of 52 cards & if the card is replaced, the result of the second draw is independent of the first. But if the card is not replaced then the result of the second depends on the result of the first draw.

Mathematical (Puriori) definition of probability:

If the outcome of a trial consists n exhaustive, mutually exclusive, equally cases, of which m of them are favourable cases to an event E , then the probability of the happening of the event E , usually denoted by $P(E)$ (or) simply p is defined to be equal to m/n .

$$\text{ie., } P(E) = P = \frac{\text{no of favourable cases}}{\text{no of possible cases}} = \frac{m}{n}$$

The probability can atmost be equal to 1, $\therefore$ the no of favourable cases & the no of possible cases can atmost coincide with each other.

Since m cases are favourable to the event, it follows that $(n-m)$ cases are not favourable to the event.

This set of unfavourable events is denoted by $\bar{E}$ or E' .
 $\therefore$ probability of the non happening of the event usually denoted by q is given by

$$q = \frac{n-m}{n} \quad (\text{or}) \quad P(\bar{E}) = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(E)$$

$$\text{i.e., } q = 1 - \frac{m}{n} = 1 - P \quad (\text{or}) \quad P + q = 1.$$

Equivalently $P(E) + P(\bar{E}) = 1$.

P is also referred to as the probability of success & q as the probability of failure.

Their sum is always equal to 1.

If $P(E) = 1$, E is called a sure event & if $P(E) = 0$, E is called an impossible event.

Examples :

1. The probability of getting a 'head' in tossing of coin

The possible outcomes are head & tail.

No of possible (exhaustive) cases $(n) = 2$.

No of favourable cases $(m) = 1$.

$$\therefore \text{Probability of getting head } (p) = \frac{m}{n} = \frac{1}{2}$$

2. The probability of getting (a) king (b) king or queen, when a card is drawn at random from a pack of 52 cards.

No of possible (exhaustive) cases $(n) = 52$

(a) No of favourable cases $(m) = 4$

$$\therefore \text{Probability of getting a king } (P) = \frac{m}{n} = \frac{4}{52} = \frac{1}{13}$$

(b) No of favourable cases (m) = $4 + 4 = 8$.

∴ Probability of getting a king (or) queen (P) =

$$\frac{m}{n} = \frac{8}{52} = \frac{2}{13}$$

3. The probability of getting (a) a no greater than 2.

(b) an odd no (c) an even no when a 'die' is thrown.

No of possible cases/outcomes (n) = 6.

(a) No of favourable outcomes = (m) = 4. Since 3, 4, 5, 6 nos are favourable to the event.

∴ Probability of getting a no greater than

$$2(P) = \frac{4}{6} = \frac{2}{3}$$

(b) No of favourable outcomes (m) = 3. Since 1, 3, 5 are favourable to the event.

∴ Probability of getting an odd no (P) = $\frac{3}{6} = \frac{1}{2}$

(c) We have (P) = $\frac{3}{6} = \frac{1}{2}$, since 2, 4, 6 are favourable to the event.

• Addition theorem of probability.

The probability of the happening of one (or) the other mutually exclusive events is equal to the sum of the probabilities of the two events.

ie., If A, B are two mutually exclusive events

then,

$$P(A \text{ or } B) = P(A) + P(B)$$

• Product theorem of Probability

If a compound event is made up of a no of independent events, the probability of the happening of

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of the compound event is equal to the product of the probabilities of the independent events.

ie., If A & B are independent events then
 $P(A \text{ and } B) = P(A) \cdot P(B)$.

Random Variables

In a random experiment, if a real variable is associated with every outcome then it is called a random variable (or) stochastic variable. In other words a random variable is a function that assigns a real no to every sample point in the sample space of a random experiment. Random variables are usually denoted by $x, y, z, \dots$ and it may be noted that different random variables may be associated with the same sample space S . The set of all real nos of a random variable x is called the range of x .

Ex: While tossing a coin, suppose that the value 1 is associated for the outcome 'head' and 0 for the outcome 'tail'. We have the sample space.

$S = \{H, T\}$ & if x is the random variable then
 $x(H) = 1$ and $x(T) = 0$ Range of $x = \{0, 1\}$

Ex: Suppose a coin is tossed twice, we shall associate two different random variables x, y as follows where we have the sample space

$S = \{HH, HT, TH, TT\}$

$x = \text{No of 'heads' in the outcome.}$

The association of the elements in S to x is as follows.

Out come	HH	HT	TH	TT
Random Variable x	2	1	1	0

Range of $x = \{0, 1, 2\}$

Suppose $Y =$ No of 'tails' in the outcome.

Out come	HH	HT	TH	TT
Random variable Y	0	1	1	2

Range of $Y = \{0, 1, 2\}$

Discrete & Continuous Random Variables

If a random variable takes finite (or) countably infinite no of values then it is called a discrete random variable. Here countably infinite means a sequence of real nos. It is evident that a discrete random variable will have finite (or) countably infinite range.

If a random variable takes non countable infinite no of values then it is called a non discrete (or) continuous random variable. Equivalently we can say that, if the range of a random variable x is an interval of real nos then x is a continuous random variable. A continuous random variable can assume any value in the interval of real nos.

Example - 1.

(a) Tossing a coin & observing the outcome.

(b) Tossing coins & observing the no of heads turning up.

(c) Throwing a 'die' & observing the no's on the face.

These are some of the examples of a discrete random variable.

Example - 2:

(a) Weight of articles.

(b) Observing the pointer on a voltmeter.

(c) Conducting a survey on the life of electric bulbs.

These are some of the examples of a continuous random variable.

Generally counting problems correspond to discrete random variables & measuring problems lead to continuous random variables.

According to the category of the random variable we have two types of probability distributions.

• Probability function & Discrete probability distribution.

If for each value x_i of a discrete random variable x , we assign a real no $P(x_i) \geq 0$

$$(1) P(x_i) \geq 0$$

$$(2) \sum_i P(x_i) = 1$$

then the funcⁿ $P(x)$ is called a probability function.

If the probability that x take the values x_i is P_i then

$$P(x = x_i) = P_i \quad (\text{or}) \quad P(x_i)$$

The set of values $[x_i, P(x_i)]$ is called a discrete (finite) probability distribution of the discrete random variable x . The function $P(x)$ is called the probability density function (p.d.f) or the probability mass function (p.m.f)

The distribution funcⁿ $F(x)$ defined by $F(x) = P(X \leq x) = \sum_{i=1}^x P(x_i)$, x being an integer is called the cumulative distribution funcⁿ (c.d.f)

The definitions of mean ($\bar{x}$) & variance (σ^2) for a general frequency distribution. i.e.,

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \quad \& \quad \sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}$$

We can as well define the mean & variance of the discrete probability distribution. In this case $P(x_i)$ corresponds to f_i & we have $\sum f_i = \sum P(x_i) = 1$.

The mean & variance of the discrete probability distribution is defined as follows.

$$\text{Mean } (\mu) = \sum_i x_i \cdot p(x_i) \quad \text{--- (1)}$$

$$\text{Variance } (V) = \sum_i (x_i - \mu)^2 \cdot p(x_i) \quad \text{--- (2)}$$

$$\text{Standard deviation } (\sigma) = \sqrt{V} \quad \text{--- (3)}$$

Note: Variance can also be put in the form

$$V = \sum_i x_i^2 p(x_i) - \left[\sum_i x_i p(x_i) \right]^2$$

Problems

1) A coin is tossed twice. A random variable X represents the no of heads turning up. Find the discrete probability distribution for X . Also find its mean & variance.

$S = \{HH, HT, TH, TT\}$. The association of the elements of S to the random variable X are respectively 2, 1, 1, 0.

$$\text{Now, } P(HH) = \frac{1}{4}, P(HT) = \frac{1}{4}, P(TH) = \frac{1}{4}, P(TT) = \frac{1}{4}$$

$$P(X=0, \text{ i.e., no head}) = P(TT) = \frac{1}{4}$$

$$P(X=1, \text{ i.e., 1 head}) = P(HT \cup TH) = P(HT) + P(TH) = \frac{1}{2}$$

$$P(X=2, \text{ i.e., 2 heads}) = P(HH) = \frac{1}{4}$$

The discrete probability distribution for X is as follows

$X = x_i$	0	1	2
$P(x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

We observe that $P(x_i) > 0$ & $\sum P(x_i) = 1$

$$\text{mean} = \mu = \sum x_i \cdot P(x_i) = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

$$\text{variance} = V = \sum (x_i - \mu)^2 \cdot P(x_i)$$

$$V = (0-1)^2 \cdot \frac{1}{4} + (1-1)^2 \cdot \frac{1}{2} + (2-1)^2 \cdot \frac{1}{4} = \frac{1}{2}$$

Thus we have mean = 1 & Variance = $\frac{1}{2}$.

∴

2) Find the value of k if the following distribution represents a finite probability distribution. Hence find its mean & standard deviation. Also find $p(x \leq 1)$, $p(x > 1)$ & $p(-1 < x \leq 2)$

x	-3	-2	-1	0	1	2	3
$p(x)$	k	$2k$	$3k$	$4k$	$3k$	$2k$	k

We must have $p(x) \geq 0$ for all x & $\sum p(x) = 1$. The first condition is satisfied if $k \geq 0$ & the second condition requires that,

$$k + 2k + 3k + 4k + 3k + 2k + k = 1 \quad (\text{or}) \quad 16k = 1 \quad \therefore k = \frac{1}{16}$$

The discrete / finite probability distribution is as follows.

x	-3	-2	-1	0	1	2	3
$p(x)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

$$\text{Mean} = \mu = \sum x \cdot p(x) = \frac{1}{16} (-3 - 4 - 3 + 0 + 3 + 4 + 3) = 0$$

$$\text{Variance} = \sigma^2 = \sum (x - \mu)^2 \cdot p(x)$$

$$= \frac{1}{16} (9 + 8 + 3 + 0 + 3 + 8 + 9) = \frac{40}{16} = \frac{5}{2}$$

$$\text{Thus } k = \frac{1}{16}, \quad \text{Mean} = 0 \quad \& \quad \text{S.D} = \sqrt{\frac{5}{2}}$$

$$\text{Also, } p(x \leq 1) = p(-3) + p(-2) + p(-1) + p(0) + p(1) = \frac{13}{16}$$

$$p(x > 1) = p(2) + p(3) = \frac{3}{16}$$

$$p(-1 < x \leq 2) = p(0) + p(1) + p(2) = \frac{9}{16}$$

3) Show that the following distribution represents a discrete probability distribution. Find the mean & variance.

x	10	20	30	40
$p(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

We observe that $p(x) > 0$ for all x & $\sum p(x) = 1$

$$\text{Mean} = \mu = \sum p(x) \cdot x = \frac{10 + 60 + 90 + 40}{8} = 25$$

$$\text{Variance} = V = \sum (x - \mu)^2 \cdot p(x)$$

$$= 225 \cdot \frac{1}{8} + 25 \cdot \frac{3}{8} + 25 \cdot \frac{3}{8} + 225 \cdot \frac{1}{8}$$

$$V = \frac{600}{8} = 75$$

Thus Mean = 25 & Variance = 75.

4) The probability distribution of a finite random variable x is given by the following table.

x_i	-2	-1	0	1	2	3
$P(x_i)$	0.1	k	0.2	$2k$	0.3	k

Find the value of k , mean & variance.

We must have $P(x_i) \geq 0$ & $\sum P(x_i) = 1$ for a probability distribution.

$$\sum P(x_i) = 1 \text{ requires } 4k + 0.6 = 1 \quad \therefore k = 0.1$$

$$\text{Mean } (\mu) = \sum x_i P(x_i) = -0.2 - 0.1 + 0.2 + 0.6 + 0.3 = 0.8$$

$$\text{Variance } (\sigma^2) = \sum x_i^2 P(x_i) - \mu^2$$

$$\sigma^2 = (0.4 + 0.1 + 0.2 + 1.2 + 0.9) - (0.8)^2 = 2.16$$

Thus Mean = 0.8 & Variance = 2.16

5) A random variable X has the following probability funcⁿ for various values of x .

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

(i) Find k (ii) Evaluate $P(x < 6)$, $P(x \geq 6)$ & $P(3 < x \leq 6)$

Also find the probability distribution & the distribution funcⁿ of X .

We must have $P(x) \geq 0$ & $\sum P(x) = 1$.

The first condition is satisfied for $k \geq 0$ & We have to find k $\exists$ $\sum P(x) = 1$.

$$\text{i.e., } 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + (7k^2 + k) = 1$$

$$\text{i.e., } 10k^2 + 9k - 1 = 0 \quad (\text{or}) \quad (10k - 1)(k + 1) = 0 \quad (\text{or}) \quad k = \frac{1}{10} \text{ \& } k = -1$$

If $k = -1$ the first condition fails & hence $k \neq -1$

$$\therefore k = +\frac{1}{10}$$

Hence We have the following table.

x	0	1	2	3	4	5	6	7
$P(x)$	0	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{1}{50}$	$\frac{17}{100}$

$$\text{Now } P(x < 6) = P(0) + P(1) + P(2) + P(3) + P(4) + P(5)$$

$$\text{i.e., } P(x < 6) = 0 + \frac{1}{10} + \frac{1}{5} + \frac{1}{5} + \frac{3}{10} + \frac{1}{100} = 0.81$$

$$P(x \geq 6) = P(6) + P(7)$$

$$\text{i.e., } P(x \geq 6) = \frac{1}{50} + \frac{17}{100} = \frac{19}{100} = 0.19$$

$$P(3 < x \leq 6) = P(4) + P(5) + P(6)$$

$$\text{i.e., } P(3 < x \leq 6) = \frac{3}{10} + \frac{1}{100} + \frac{1}{50} = \frac{33}{100} = 0.33$$

The probability distribution is as follows.

x	0	1	2	3	4	5	6	7
$P(x)$	0	0.1	0.2	0.2	0.3	0.01	0.02	0.17

The distribution funcⁿ of X is $f(x) = P(X \leq x) = \sum_{i=1}^x P(x_i)$ is also called cumulative distribution funcⁿ & the same is as follows.

x	0	1	2	3	4	5	6	7
$f(x)$	0	$0+0.1$ $=0.1$	$0.1+0.2$ $=0.3$	$0.3+0.2$ $=0.5$	$0.5+0.3$ $=0.8$	$0.8+0.01$ $=0.81$	$0.81+0.02$ $=0.83$	$0.83+0.17$ $=1$

6) If X is a discrete random variable taking values $1, 2, 3, \dots$ with $p(x) = \frac{1}{2} \cdot \left(\frac{2}{3}\right)^x$, find $p(x \text{ being an odd number})$ by first establishing that $p(x)$ is a probability funcⁿ.

$$\begin{aligned} \sum_i P(x_i) &= \sum_{x=1}^{\infty} \frac{1}{2} \left(\frac{2}{3}\right)^x \\ &= \frac{1}{2} \left[\frac{2}{3} + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \dots \right] \\ &= \frac{1}{2} \cdot \frac{\frac{2}{3}}{1 - \left(\frac{2}{3}\right)} = \frac{1}{2} \cdot \frac{\frac{2}{3}}{\frac{1}{3}} = 1 \end{aligned}$$

$\therefore p(x)$ is a probability function.

$$\text{Next } p(X \text{ being an odd no.}) = \sum_{x=1,3,5,\dots}^{\infty} P(x) = \sum_{x=1,3,5,\dots}^{\infty} \frac{1}{2} \left(\frac{2}{3}\right)^x$$

$$= \frac{1}{2} \left[\frac{2}{3} + \left(\frac{2}{3}\right)^3 + \left(\frac{2}{3}\right)^5 + \dots \right]$$

$$= \frac{1}{2} [a + ar + ar^2 + \dots]$$

$$\text{Where } a = \frac{2}{3}, \quad r = \left(\frac{2}{3}\right)^2$$

$$= \frac{1}{2} \cdot \frac{a}{1-r} = \frac{1}{2} \cdot \frac{\frac{2}{3}}{1 - \left(\frac{2}{3}\right)^2}$$

$$= \frac{1}{2} \cdot \frac{2/3}{5/9} = \frac{3}{5}$$

Thus $P(X \text{ being an odd no}) = \frac{3}{5}$.

7) The range of a random variable $X = \{1, 2, 3, \dots\}$ & the probabilities of X are $\exists$ $P(X=k) = \frac{\lambda^k}{k!}$ Where $k = 1, 2, 3, \dots$. Find the value of λ & $P(0 < X < 3)$.

We must have $\sum P(X) = 1$

$$\text{i.e., } \sum_{k=1}^{\infty} \frac{\lambda^k}{k!} = 1$$

$$(\text{or}) \quad \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots = 1$$

$$\text{i.e., } (e^\lambda - 1) = 1 \quad (\text{or}) \quad e^\lambda = 2 \Rightarrow \lambda = \log_e 2$$

$$\text{Also } P(0 < X < 3) = P(X=1) + P(X=2)$$

$$= \frac{\lambda}{1!} + \frac{\lambda^2}{2!} \quad \text{Where } \lambda = \log_e 2$$

$$\text{Thus } P(0 < X < 3) = \log_e 2 + \frac{1}{2} (\log_e 2)^2$$

Repeated Trails:

A random experiment with only two possible outcomes categorized as success & failure is called a Bernoulli trial. Where the probability of success p is same for each trial.

Bernoulli's theorem.

A random experiment with only two possible outcomes cat.

The probability of x successes in n trials is equal to ${}^n C_x p^x q^{n-x}$ where p is the probability of success & q is the probability of failure.

Binomial Distribution.

If p is the probability of success & q is the probability of failure, the probability of x successes out of n trials is given by

$$P(x) = {}^n C_x p^x q^{n-x}$$

We form the following probability distribution of $[x, p(x)]$ Where $x = 0, 1, 2, \dots, n$.

x	0	1	2	...	n
$P(x)$	q^n	${}^n C_1 q^{n-1} p$	${}^n C_2 q^{n-2} p^2$	...	p^n

It may be observed that the value of $P(x)$ for different values $x = 0, 1, 2, \dots, n$ are the successive terms in the binomial expansion of $(q+p)^n$ & accordingly this distribution is called the Binomial Distribution or Bernoulli Distribution.

$$\sum P(x) = q^n + {}^n C_1 q^{n-1} p + {}^n C_2 q^{n-2} p^2 + \dots + p^n = (q+p)^n = 1^n = 1$$

Hence $P(x)$ is a probability funcⁿ.

• Mean & Standard Deviation of the Binomial Distribution

$$\text{Mean } (\mu) = \sum_{x=0}^n x p(x)$$

$$\mu = \sum_{x=0}^n x \cdot {}^n C_x p^x q^{n-x}$$

$$= \sum_{x=0}^n x \cdot \frac{n!}{x! (n-x)!} p^x q^{n-x}$$

$$= \sum_{x=0}^n \frac{n \cdot (n-1)!}{(x-1)! (n-x)!} p \cdot p^{x-1} q^{n-x}$$

$$= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)! [(n-1)-(x-1)]!} \cdot p^{x-1} q^{(n-1)-(x-1)}$$

$$= np \sum_{x=1}^n {}^{(n-1)}C_{(x-1)} p^{x-1} q^{(n-1)-(x-1)}$$

$$\mu = np (q+p)^{n-1} = np$$

$$\text{Mean}(\mu) = np$$

$$\text{Variance} (V) = \sum_{x=0}^n x^2 p(x) - \mu^2 \quad \text{--- (1)}$$

$$\text{Now } \sum_{x=0}^n x^2 p(x) = \sum_{x=0}^n [x(x-1) + x] p(x)$$

$$= \sum_{x=0}^n x(x-1) p(x) + \sum_{x=0}^n x p(x)$$

$$= \sum_{x=0}^n x(x-1) {}^nC_x p^x q^{n-x} + np$$

$$= \sum_{x=0}^n x(x-1) \frac{n!}{x! (n-x)!} p^x q^{n-x} + np$$

$$= \sum_{x=0}^n \frac{n(n-1) \cdot (n-2)!}{(x-2)! (n-x)!} p^2 p^{x-2} q^{n-x} + np$$

$$= n(n-1) p^2 \sum_{x=2}^n \frac{(n-2)!}{(x-2)! [(n-2)-(x-2)]!} p^{x-2} q^{(n-2)-(x-2)} + np$$

$$= n(n-1) p^2 \sum_{x=2}^n {}^{(n-2)}C_{(x-2)} p^{x-2} q^{(n-2)-(x-2)} + np$$

$$= n(n-1) p^2 (q+p)^{n-2} + np$$

$$\therefore \sum x^2 p(x) = n(n-1)p^2 + np$$

Using this result in (1) along with $\mu = np$ We have

$$\text{Variance (V)} = \{n(n-1)p^2 + np\} - (np)^2$$

$$= n^2 p^2 - np^2 + np - n^2 p^2 = np(1-p) = npq$$

Hence Variance (V) = npq.

$$\text{S.D}(\sigma) = \sqrt{V} = \sqrt{npq}$$

Thus We have for the Binomial Distribution,

$$\text{Mean } (\mu) = np \quad \& \quad \text{S.D}(\sigma) = \sqrt{npq}$$

Poisson Distribution

Poisson distribution is regarded as the limiting form of the binomial distribution when n is very large ($n \rightarrow \infty$) & p the probability of success is very small ($p \rightarrow 0$) so that np tends to a fixed constant say m .

$$\therefore P(x) = \frac{m^x e^{-m}}{x!}$$

Mean & Standard Deviation of the Poisson Distribution

$$\text{Mean } (\mu) = \sum_{x=0}^{\infty} x \cdot P(x)$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{m^x e^{-m}}{x!} = \sum_{x=1}^{\infty} \frac{m^x e^{-m}}{(x-1)!}$$

$$\mu = m e^{-m} \sum_{x=1}^{\infty} \frac{m^{x-1}}{(x-1)!}$$

$$= m e^{-m} \left[1 + \frac{m}{1!} + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots \right]$$

$$= m e^{-m} \cdot e^m = m \cdot e^0 = m$$

$$\therefore \text{Mean } (\mu) = m.$$

$$\text{Variance } (V) = \sum_{x=0}^{\infty} x^2 P(x) - \mu^2 \quad \text{--- (1)}$$

$$\begin{aligned} \text{Now } \sum x^2 P(x) &= \sum [x(x-1) + x] \frac{m^x e^{-m}}{x!} \\ &= \sum \frac{m^x e^{-m}}{(x-2)!} + \sum \frac{m^x e^{-m}}{(x-1)!} \end{aligned}$$

$$\begin{aligned} \sum x^2 P(x) &= m^2 e^{-m} \sum_{x=2}^{\infty} \frac{m^{x-2}}{(x-2)!} + m \\ &= m^2 e^{-m} \left[1 + \frac{m}{1!} + \frac{m^2}{2!} + \dots \right] + m \\ &= m^2 e^{-m} \cdot e^m + m \end{aligned}$$

Using this result in (1) along with $\mu = m$ we have,

$$\text{Variance } (V) = m^2 + m - (m)^2 = m$$

$$\therefore \text{S.D } (\sigma) = \sqrt{V} = \sqrt{m}$$

Thus we have for the Poisson distribution

$$\text{Mean } (\mu) = m \quad \& \quad \text{S.D } (\sigma) = \sqrt{m}$$

Further we can say that the mean & variance are equal for the Poisson distribution.

Problems.

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1) Find the binomial probability distribution which has mean 2 & variance $\frac{4}{3}$.

WKT for the binomial distribution mean = np & Variance = npq .

Hence $np = 2$ & $npq = \frac{4}{3}$ by using the data.

Further $2q = \frac{4}{3}$ (or) $q = \frac{2}{3} \therefore p = 1 - q = \frac{1}{3}$

Since $np = 2$, We have $n(\frac{1}{3}) = 2 \therefore n = 6$

The binomial probability function $P(x) = {}^nC_x p^x q^{n-x}$ becomes $P(x) = {}^6C_x (\frac{1}{3})^x (\frac{2}{3})^{6-x}$

The distribution of probabilities is as follows.

x	0	1	2	3
$P(x)$	$(\frac{2}{3})^6$	${}^6C_1 (\frac{1}{3}) (\frac{2}{3})^5$	${}^6C_2 (\frac{1}{3})^2 (\frac{2}{3})^4$	${}^6C_3 (\frac{1}{3})^3 (\frac{2}{3})^3$
		4	5	6
		${}^6C_4 (\frac{1}{3})^4 (\frac{2}{3})^2$	${}^6C_5 (\frac{1}{3})^5 (\frac{2}{3})$	$(\frac{1}{3})^6$

2) When a coin is tossed 4 times, find the probability of getting (i) exactly one head (ii) atmost 3 heads (iii) atleast 2 heads.

$$p = P(H) = \frac{1}{2} = 0.5, \quad q = 0.5; \quad n = 4$$

WKT the probability of x successes out of n trials is given by $P(x) = {}^nC_x p^x q^{n-x}$

$$(i) P(1 \text{ head}) = P(x=1) = {}^4C_1 (0.5)^1 (0.5)^3 = 0.25$$

$$(ii) P(\text{atmost 3 heads}) = P(x=0) + P(x=1) + P(x=2) + P(x=3) \\ = {}^4C_0 (0.5)^0 (0.5)^4 + {}^4C_1 (0.5)^1 (0.5)^3 + {}^4C_2 (0.5)^2 (0.5)^2 + {}^4C_3 (0.5)^3 (0.5)^1$$

$$= (0.5)^4 (1+4+6+4) = 0.9375$$

$$\text{(iii) } P(\text{atleast 2 heads}) = 1 - [P(X=0) + P(X=1)]$$

$$= 1 - [(0.5)^4 + {}^4C_1 (0.5)(0.5)^3] = 0.6875$$

3) The probability that a person aged 60 years will live upto 70 is 0.65. What is the probability that out of 10 persons aged 60 atleast 7 of them will live upto 70.

Let x be the no of persons aged 60 years living upto 70 years.

For this variate we have by data

$$p = 0.65. \quad \text{Hence } q = 0.35$$

$$\text{Consider } P(x) = {}^nC_x p^x q^{n-x}. \quad \text{Hence } n = 10$$

We have to find $P(x \geq 7)$. i.e., given by

$$= P(7) + P(8) + P(9) + P(10)$$

$$= {}^{10}C_7 (0.65)^7 (0.35)^3 + {}^{10}C_8 (0.65)^8 (0.35)^2 + {}^{10}C_9 (0.65)^9 (0.35) + (0.65)^{10}$$

$$\text{But } {}^{10}C_7 = {}^{10}C_3 = \frac{10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3} = 120; \quad {}^{10}C_8 = {}^{10}C_2 = \frac{10 \cdot 9}{1 \cdot 2} = 45;$$

$${}^{10}C_9 = {}^{10}C_1 = 10$$

$$\text{Hence } P(x \geq 7) = 0.5138.$$

4) The no of telephone lines busy at an instant of time is a binomial variate with probability 0.1 that a line is busy. If 10 lines are chosen at random, what is the probability that (i) no line is busy (ii) all lines are busy (iii) atleast one line is busy (iv) atmost 2 lines are busy.

Let x denote the no of telephone lines busy.

For this variate we have by data,

$$P = 0.1 ; q = 1 - P = 0.9 . \text{ Also } n = 10$$

$$\text{We have } P(x) = {}^n C_x P^x q^{n-x} = {}^{10} C_x (0.1)^x (0.9)^{10-x}$$

$$(i) \text{ Probability that no line is busy } = P(0) = (0.9)^{10} = 0.3487$$

$$(ii) \text{ Probability that all lines are busy } = P(10) = (0.1)^{10}$$

$$(iii) \text{ Probability that atleast one line is busy} \\ = 1 - \text{Probability of no line is busy} \\ = 1 - P(0) = 1 - 0.3487 = 0.6513$$

$$(iv) \text{ Probability that atmost 2 lines are busy}$$

$$= P(0) + P(1) + P(2)$$

$$= (0.9)^{10} + {}^{10} C_1 (0.1)^1 (0.9)^9 + {}^{10} C_2 (0.1)^2 (0.9)^8$$

$$= 0.9298.$$

5) In a quiz contest of answering 'Yes' or 'No' what is the probability of guessing atleast 6 answers correctly out of 10 questions asked? Also find the probability of the same if there are 4 options for a correct answer.

Let x denote the correct answer & We have in the first case,

$$P = \frac{1}{2} \text{ \& } q = \frac{1}{2}$$

$$P(x) = {}^n C_x P^x q^{n-x} = {}^{10} C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x} = {}^{10} C_x \cdot \frac{1}{2^{10}}$$

We have to find $P(x \geq 6)$

$$P(x \geq 6) = \frac{1}{2^{10}} [{}^{10} C_6 + {}^{10} C_7 + {}^{10} C_8 + {}^{10} C_9 + {}^{10} C_{10}]$$

$$= \frac{1}{2^{10}} (210 + 120 + 45 + 10 + 1) = \frac{386}{1024} = 0.377$$

$$\text{Thus } P(x \geq 6) = 0.377.$$

In the second case when there are 4 options

$$p = 1/4, \quad q = 3/4; \quad n = 10$$

$$P(x) = {}^{10}C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{10-x} = \frac{1}{4^{10}} [3^{10-x} \cdot {}^{10}C_x]$$

$$\text{Hence } P(x \geq 6) = P(6) + P(7) + P(8) + P(9) + P(10)$$

$$= \frac{1}{4^{10}} [3^4 \cdot {}^{10}C_6 + 3^3 \cdot {}^{10}C_7 + 3^2 \cdot {}^{10}C_8 + 3 \cdot {}^{10}C_9 + {}^{10}C_{10}]$$

$$= \frac{1}{4^{10}} [81 \times 210 + 27 \times 120 + 9 \times 45 + 3 \times 10 + 1] = 0.019$$

$$\text{Thus } P(x \geq 6) = 0.01972 \approx 0.02$$

6) If the mean & standard deviation of the no correctly answered questions in a test given to 4096 students are 2.5 & $\sqrt{1.875}$. Find an estimate of the no of candidates answering correctly (i) 8 or more questions (ii) 2 or less (iii) 5 questions.

We have mean $(\mu) = np$ & SD $(\sigma) = \sqrt{npq}$ for a binomial distribution.

$$\text{By data } np = 2.5 \quad \& \quad \sqrt{npq} = \sqrt{1.875} \quad \text{or } npq = 1.875$$

$$\text{Hence we have } 2.5q = 1.875 \quad \therefore q = 0.75; \quad p = 1 - q = 0.25$$

$$\text{Since } np = 2.5 \quad \text{We have } (0.25)n = 2.5 \quad \therefore n = 10$$

Let x denote the no of correctly answered questions.

$$P(x) = {}^nC_x p^x q^{n-x} = {}^{10}C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{10-x}$$

$$\text{i.e., } P(x) = \frac{1}{4^{10}} [{}^{10}C_x (3)^{10-x}]$$

Since the estimate is needed for 4096 students

We have

$$4096 P(x) = \frac{4096}{4^{10}} [{}^{10}C_x (3)^{10-x}] = \frac{2^{12}}{2^{20}} [{}^{10}C_x (3)^{10-x}]$$

$$\text{i.e., } 4096 P(x) = \frac{1}{2^8} [{}^{10}C_x (3)^{10-x}] = f(x) \text{ (say)}$$

(i) We have to find $f(8) + f(9) + f(10)$

$$= \frac{1}{256} [{}^{10}C_8 \cdot 3^2 + {}^{10}C_9 \cdot 3 + 1]$$

$$= \frac{1}{256} [9 \cdot {}^{10}C_2 + 3 \cdot {}^{10}C_1 + 1]$$

$$= \frac{1}{256} (436) = 1.703 \approx 2.$$

No of students correctly answering 8 or more questions is 2.

(ii) We have to find $f(2) + f(1) + f(0)$

$$= \frac{1}{256} [{}^{10}C_2 \cdot 3^8 + {}^{10}C_1 \cdot 3^9 + 3^{10}]$$

$$= \frac{3^8}{256} (45 + 30 + 9) = \frac{3^8}{256} (84) = 2152.8 \approx 2153.$$

No of students correctly answering 2 or less than 2 questions is 2153.

(iii) We have to find $f(5)$

$$= \frac{1}{256} [{}^{10}C_5 \cdot 3^5] = 239.2 \approx 239.$$

No of students correctly answering 5 questions is 239.

7) In 800 families with 5 children each how many families would be expected to have (i) 3 boys (ii) 5 girls (iii) either 2 or 3 boys (iv) at most 2 girls by assuming probabilities for boys & girls to be equal.

P = Probability of having a boy = $\frac{1}{2}$

q = Probability of having a girl = $\frac{1}{2}$

Let x denote the no of boys in the family.

$$P(x) = {}^nC_x p^x q^{n-x} \quad \text{Where } n=5.$$

$$\text{ie., } P(x) = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x} = \frac{1}{2^5} \cdot {}^5C_x = \frac{{}^5C_x}{32}$$

Since we have to find the expected no in respect of 800 families we have

$$800 P(x) = 800 \cdot \frac{{}^5C_x}{32} = 25 \cdot {}^5C_x = f(x) \text{ (say)}$$

(i) We have to find $f(3)$

$$f(3) = 25 \cdot {}^5C_3 = 25 \times 10 = 250.$$

Expected no of families with 3 boys is 250.

(ii) We have to find $f(0)$

$$f(0) = 25 \cdot {}^5C_0 = 25 \times 1 = 25$$

Expected no of families with 5 girls is 25.

(iii) We have to find $f(2) + f(3)$

$$= 25 \cdot {}^5C_2 + 25 \cdot {}^5C_3 = 50 \cdot {}^5C_2 = 50 \times 10 = 500.$$

Expected no of families with 2 or 3 boys is 500.

(iv) At most 2 girls means that, families can have 5 boys & 0 girls (or) 4 boys & 1 girl (or) 3 boys & 2 girls.

Hence we have to find $f(5) + f(4) + f(3)$

$$= 25 \cdot {}^5C_5 + 25 \cdot {}^5C_4 + 25 \cdot {}^5C_3$$

$$= 25 (1 + 5 + 10) = 25 \times 16 = 400$$

Expected no of families with atmost 2 girls is 400.

PD-13

8) Five dice were thrown 96 times & the no of times an odd no actually turned out in the experiment is given. Fit a binomial distribution to this data & calculate the expected frequencies.

No. of dice showing 1 or 3 or 5	0	1	2	3	4	5
Observed frequency	1	10	24	35	18	8

P = Probability of getting 1 or 3 or 5 = $\frac{3}{6} = \frac{1}{2} \therefore q = \frac{1}{2}$

Here let x denote the no of times an odd no turning out.

$$P(x) = {}^nC_x p^x q^{n-x} \quad \text{Where } n=5$$

$$\text{Hence } P(x) = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x} = \frac{1}{2^5} {}^5C_x$$

This is the binomial probability distribution function.

Since 5 dice were thrown 96 times, expected frequencies are obtained from $f(x) = 96 P(x)$ where $x=0, 1, 2, 3, 4, 5$

$$\text{Consider } f(x) = 96 \cdot \frac{1}{2^5} {}^5C_x = 3 \cdot {}^5C_x$$

$$\text{Hence } f(0) = 3 \cdot {}^5C_0 = 3 \quad ; \quad f(1) = 3 \cdot {}^5C_1 = 3 \times 5 = 15$$

$$f(2) = 3 \cdot {}^5C_2 = 3 \times 10 = 30 \quad ; \quad f(3) = 3 \cdot {}^5C_3 = 3 \times 10 = 30$$

$$f(4) = 3 \cdot {}^5C_4 = 3 \times 5 = 15 \quad ; \quad f(5) = 3 \cdot {}^5C_5 = 3 \times 1 = 3$$

The expected (theoretical) frequencies are 3, 15, 30, 30, 15, 3.

9) 4 coins are tossed 100 times & the following results were obtained. Fit a binomial distribution for the data & calculate the theoretical frequencies.

No of heads	0	1	2	3	4
Frequency	5	29	36	25	5

Let x denote the no of heads & f the corresponding frequency. Since the data is in the form of a frequency

distribution We shall first calculate the mean.

$$\text{Mean } (\mu) = \frac{\sum fx}{\sum f} = \frac{0+29+72+75+20}{100} = \frac{196}{100} = 1.96$$

But $\mu = np$ for the binomial distribution. Here $n=4$

$$\text{Hence } 4p = 1.96 \text{ (or) } p = 0.49 \quad \therefore q = 1-p = 0.51$$

Binomial distribution probability funcⁿ is given by

$$P(x) = {}^nC_x p^x q^{n-x} = {}^4C_x (0.49)^x (0.51)^{4-x}$$

Since 4 coins were tossed 100 times, expected (theoretical) frequencies are obtained from

$$F(x) = 100 P(x) = 100 \cdot {}^4C_x (0.49)^x (0.51)^{4-x}$$

Where $x = 0, 1, 2, 3, 4$.

$$F(0) = 100 (0.51)^4 = 6.765 \approx 7$$

$$F(1) = 100 \cdot {}^4C_1 (0.49) (0.51)^3 = 400 (0.49) (0.51)^3 = 25.999 \approx 26$$

$$F(2) = 100 \cdot {}^4C_2 (0.49)^2 (0.51)^2 = 600 (0.49)^2 (0.51)^2 = 37.47 \approx 37$$

$$F(3) = 100 \cdot {}^4C_3 (0.49)^3 (0.51) = 400 (0.49)^3 (0.51) = 24.0004 \approx 24$$

$$F(4) = 100 \cdot {}^4C_4 (0.49)^4 = 100 (0.49)^4 = 5.765 \approx 6$$

Thus the required theoretical frequencies are 7, 26, 37, 24, 6.

10) A lot contains 1% of defective items. What should be the no (n) of items in a random sample so that the probability of finding atleast one defective in it is atleast 0.75?

Let p be the probability of a defective item.

$$\text{By data } p = 1\% = 0.01 \quad \therefore q = 0.99$$

If x denotes a defective item,

$$P(x) = {}^nC_x p^x q^{n-x} = {}^nC_x (0.01)^x (0.99)^{n-x}$$

We need to find n s.t. the probability of finding atleast one defective is ≥ 0.75 .

ie., to find n s.t.

$$P(X \geq 1) \geq 0.75$$

$$\text{i.e., } 1 - P(X < 1) \geq 0.75$$

$$\text{i.e., } 1 - P(0) \geq 0.75$$

$$\text{i.e., } 1 - (0.99)^n \geq 0.75 \quad \text{or} \quad 0.25 \geq (0.99)^n$$

Equivalently We have $\log(0.25) \geq n \log(0.99)$

$$\text{(or)} \quad n \leq \frac{\log(0.25)}{\log(0.99)} = 137.935$$

Hence the required n is 138.

11) Fit a poisson distribution for the following data & calculate the theoretical frequencies:

x	0	1	2	3	4
f	122	60	15	2	1

We shall first compute the mean (μ) of the given distribution.

$$\mu = \frac{\sum fx}{\sum f} = \frac{0 + 60 + 30 + 6 + 4}{200} = 0.5$$

We have mean (μ) = m for the poisson distribution.

The Poisson distribution is

$$P(x) = \frac{m^x e^{-m}}{x!} \quad \& \quad \text{let } f(x) = 200 \cdot P(x)$$

$$\text{i.e., } f(x) = 200 \cdot \frac{(0.5)^x e^{-0.5}}{x!} \quad \text{But } e^{-0.5} \approx 0.6065.$$

$$\therefore f(x) = 121.3 \frac{(0.5)^x}{x!}$$

Putting $x = 0, 1, 2, 3, 4$ in $f(x)$ we obtain the theoretical frequencies. They are as follows.

$$121.3 \frac{(0.5)^0}{0!} \approx 121, \quad 121.3 \frac{(0.5)^1}{1!} \approx 61, \quad 121.3 \frac{(0.5)^2}{2!} \approx 15,$$

$$121.3 \frac{(0.5)^3}{3!} \approx 3, \quad 121.3 \frac{(0.5)^4}{4!} \approx 0$$

Thus the required theoretical frequencies are 121, 61, 15, 3, 0.

- 12) The no of accidents in a year to taxi drivers in a city follows a Poisson distribution with mean 3. Out of 1000 taxi drivers find approximately the no of the drivers with
- no accident in a year.
 - more than 3 accidents in a year.

By data, mean $(\mu) = 3$ & We have for the Poisson distribution $\mu = m = 3$.

The Poisson distribution is given by $P(x) = \frac{m^x e^{-m}}{x!} = \frac{e^{-3} m^x}{x!}$

Let $f(x) = 1000 P(x)$

ie., $f(x) = 1000 \frac{3^x e^{-3}}{x!} = 50 \cdot \frac{3^x}{x!}$ Since $e^{-3} = 0.05$

(i) No of drivers with no accident in a year $= f(0) = 50 \cdot \frac{3^0}{0!} = 50$

(ii) Probability of more than 3 accidents in a year $= 1 - P(x \leq 3)$

ie., $= 1 - [P(0) + P(1) + P(2) + P(3)]$

$$= 1 - \left[e^{-3} \left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!} + \frac{3^3}{3!} \right) \right]$$

$$= 1 - [0.05(1 + 3 + 4.5 + 4.5)] = 0.35$$

$\therefore$ no of drivers out of 1000 With more than 3 accidents in a year $= 1000 \times 0.35 = 350$.

- 13) 2% of the fuses manufactured by a firm are found to be defective. Find the probability that a box containing 200 fuses contains (i) no defective fuses (ii) 3 or more defective fuses.

$P =$ Probability of a defective fuse $= 2/100 = 0.02$

$\therefore$ mean no of defectives $\mu = m = np = 200 \times 0.02 = 4$

The Poisson distribution is given by $P(x) = \frac{m^x e^{-m}}{x!}$

ie., $P(x) = \frac{4^x e^{-4}}{x!}$ But $e^{-4} = 0.0183$

$$\therefore P(x) = 0.0183 \cdot \frac{4^x}{x!}$$

(i) Probability of no defective fuse = $P(0) = 0.0183$

(ii) Probability of 3 (or) more defective fuses

$$= 1 - [P(0) + P(1) + P(2)]$$

$$= 1 - 0.0183 \left[1 + \frac{4^1}{1!} + \frac{4^2}{2!} \right]$$

$$= 1 - 0.0183 (1 + 4 + 8) \approx 0.7621.$$

14) A communication channel receives independent pulses at the rate of 12 pulses per micro second. The probability of transmission error is 0.001 for each micro second. Compute the probabilities of (i) no error during a micro second (ii) One error per micro second (iii) atleast one error per micro second (iv) two errors (v) atmost two errors.

Mean no of errors in one micro second.

$$\mu = np = m = 12 \times 0.001 = 0.012$$

The Poisson distribution is $P(x) = \frac{m^x e^{-m}}{x!}$

ie., $P(x) = \frac{e^{-0.012} (0.012)^x}{x!}$

(i) $P(0) = 0.988072$ (ii) $P(1) = e^{-0.012} \times 0.012 = 0.01186$

(iii) Probability of atleast one error

$$= 1 - P(0) = 1 - e^{-0.012} = 0.01193$$

(iv) $P(2) = \frac{e^{-0.012} (0.012)^2}{2!} = 0.000071$

(v) Probability of atmost two errors

$$= P(0) + P(1) + P(2)$$

$$= e^{-0.012} \left[1 + 0.012 + \frac{(0.012)^2}{2} \right] = 0.9999 \approx 1.$$

- 15) A shop has 4 diesel generator sets which it hires every day. The demand for a gen set on an average is a poisson variate with value $\frac{5}{2}$. Obtain the Probability that on a particular day (i) there was no demand (ii) a demand had to be refused.

By data m = mean demand for a generator = $\frac{5}{2} = 2.5$
 Poisson distribution is given by $P(x) = \frac{m^x e^{-m}}{x!} = e^{-2.5} \cdot \frac{(2.5)^x}{x!}$

- (i) No demand for a generator. We have to find $P(0)$

$$P(0) = e^{-2.5} = 0.082085$$

- (ii) If a demand had to be refused, there should have been a demand for more than generators.
 We have to find $P(x > 4)$

$$P(x > 4) = 1 - P(x \leq 4)$$

$$= 1 - [P(0) + P(1) + P(2) + P(3) + P(4)]$$

$$= 1 - e^{-2.5} \left[1 + 2.5 + \frac{(2.5)^2}{2!} + \frac{(2.5)^3}{3!} + \frac{(2.5)^4}{4!} \right]$$

$$\text{Thus } P(x > 4) = 0.108822.$$

- 16) The probability that a news reader commits no mistake in reading the news is $\frac{1}{e^3}$. Find the probability that on a particular news broadcast he commits.

- (i) Only 2 mistakes (ii) more than 3 mistakes

- (iii) atmost 3 mistakes.

$$\frac{1}{e^3} = e^{-3}$$

$$(i) P(2) = 0.22404$$

$$(ii) P(x > 3) = 1 - P(x \leq 3) = 0.3528$$

$$(iii) \text{ atmost } = P(0) + P(1) + P(2) + P(3) = e^{-3} (1+3) = 0.6470$$

17) If x is a poisson variate $\exists$ $P(x=2) = 9P(x=4) + 90P(x=6)$,
 Compute the mean & variance of the Poisson distribution.
 We have $P(x) = \frac{m^x e^{-m}}{x!}$ & by using the data we have,

$$\frac{m^2 e^{-m}}{2!} = 9 \frac{m^4 e^{-m}}{4!} + 90 \frac{e^{-m} m^6}{6!}$$

$$\text{ie., } \frac{m^2}{2} = \frac{9}{24} m^4 + \frac{90}{720} m^6$$

$$\text{ie., } \frac{m^2}{2} = \frac{3}{8} m^4 + \frac{1}{8} m^6 \quad (\text{or}) \quad 1 = \frac{3}{4} m^2 + \frac{1}{4} m^4$$

$$\text{ie., } m^4 + 3m^2 - 4 = 0 \quad (\text{or}) \quad (m^2 - 1)(m^2 + 4) = 0$$

$$\therefore m = \pm 1, \quad m = \pm 2i.$$

Neglecting the imaginary value of m & noting that in a Poisson distribution mean & variance are equal to m we have,
 mean = variance = 1, taking the positive value.

18) Prove the following recurrence relation for a Poisson distribution

$$P(x+1) = \frac{m}{x+1} P(x)$$

$$\Rightarrow \text{We have } P(x) = \frac{m^x e^{-m}}{x!} \quad \text{--- (1)}$$

$$\begin{aligned} \therefore P(x+1) &= \frac{m^{x+1} e^{-m}}{(x+1)!} \\ &= \frac{m \cdot m^x e^{-m}}{(x+1) \cdot x!} \end{aligned}$$

$$P(x+1) = \frac{m}{x+1} \cdot \frac{m^x e^{-m}}{x!} = \frac{m}{x+1} P(x) \quad \text{by using (1).}$$

$$\text{Thus } P(x+1) = \frac{m}{x+1} P(x).$$

///

Continuous Probability Distribution.

If for every x belonging to the range of a continuous random variable X , we assign a real no $f(x)$ satisfying the conditions

$$1) f(x) \geq 0 \quad 2) \int_{-\infty}^{\infty} f(x) dx = 1.$$

then $f(x)$ is called a continuous probability function or Probability density function (P.d.f)

If (a, b) is a subinterval of the range space of x then the probability that ' x ' lies in (a, b) is defined to be the interval of $f(x)$ b/w a & b . i.e.,

$$P(a \leq x \leq b) = \int_a^b f(x) dx. \quad \text{--- (1)}$$

Cumulative distribution function.

If x is a continuous random variable with probability density funcⁿ $f(x)$ then the funcⁿ $F(x)$ defined by

$$F(x) = P(x \leq x) = \int_{-\infty}^x f(x) dx \quad \text{--- (2)}$$

is called the cumulative distribution funcⁿ (c.d.f) of x

It is evident from (2) that

$$F(x) = P(x \leq x) = P(-\infty < x \leq x) \quad \& \quad \frac{d}{dx} [F(x)] = f(x).$$

It should be noted that the Probability of a continuous random variable taking a particular value is zero

Whereas the probability that it take values in an interval is a positive quantity.

If x is any real no then

$$P(x \geq a) = \int_a^{\infty} f(x) dx \quad \text{--- (3)}$$

$$P(x < a) = 1 - P(x \geq a)$$

$$\text{i.e., } P(x < a) = 1 - \int_a^{\infty} f(x) dx \quad \text{--- (4)}$$

Mean & Variance

If x is a continuous random variable with probability density funcⁿ $f(x)$ where $-\infty < x < \infty$, the mean (μ) or expectation $E(x)$ & the variance (σ^2) of x is defined as follows.

$$\text{Mean } (\mu) = \int_{-\infty}^{\infty} x \cdot f(x) dx \quad \text{--- (5)}$$

$$\text{Variance } (\sigma^2) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx \quad \text{--- (6)}$$

We now introduce two continuous probability distributions namely the exponential distribution & the normal distributions.

Exponential Distribution.

The continuous probability distribution having the probability density funcⁿ $f(x)$ given by

$$f(x) = \begin{cases} \alpha e^{-\alpha x} & \text{for } x > 0 \\ 0 & \text{otherwise, where } \alpha > 0 \end{cases}$$

is known as the exponential distribution.

Evidently $f(x) > 0$ & We have

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \alpha e^{-\alpha x} dx = \alpha \left[\frac{e^{-\alpha x}}{-\alpha} \right]_0^{\infty} = -(0 - 1) = 1$$

$$\text{Thus } \int_{-\infty}^{\infty} f(x) dx = 1.$$

$f(x)$ satisfy both the conditions required for a Continuous probability funcⁿ / probability density funcⁿ.

• Mean & Standard Deviation of the Exponential Distribution.

$$\text{Mean } (\mu) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_{-\infty}^{\infty} x \cdot \alpha e^{-\alpha x} dx.$$

$$\mu = \alpha \int_0^{\infty} x e^{-\alpha x} dx$$

Applying Bernoulli's rule of integration by parts we have,

$$\mu = \alpha \left[x \cdot \left(\frac{e^{-\alpha x}}{-\alpha} \right) - 1 \cdot \left(\frac{e^{-\alpha x}}{\alpha^2} \right) \right]_0^{\infty}$$

(Here $x/e^{\alpha x} \rightarrow 0$ as $x \rightarrow \infty$ by L'Hospital's rule)

$$\mu = \alpha \left[0 - \frac{1}{\alpha^2} (0 - 1) \right] = \frac{1}{\alpha} ; \mu = \frac{1}{\alpha}$$

$$\text{Variance } (\sigma^2) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx$$

$$\sigma^2 = \alpha \int_{-\infty}^{\infty} (x - \mu)^2 e^{-\alpha x} dx$$

Applying Bernoulli's rule we have,

$$\sigma^2 = \alpha \left[(x - \mu)^2 \cdot \left(\frac{e^{-\alpha x}}{-\alpha} \right) - 2(x - \mu) \cdot \left(\frac{e^{-\alpha x}}{\alpha^2} \right) + 2 \cdot \frac{e^{-\alpha x}}{-\alpha^3} \right]_0^{\infty}$$

$$= \alpha \left[\frac{-1}{\alpha} \{0 - \mu^2\} - \frac{2}{\alpha^2} \{0 - (-\mu)\} - \frac{2}{\alpha^3} \{0 - 1\} \right]$$

$$= \alpha \left[\frac{\mu^2}{\alpha} - \frac{2\mu}{\alpha^2} + \frac{2}{\alpha^3} \right] \quad \text{But } \mu = 1/\alpha$$

$$\therefore \sigma^2 = \alpha \left[\frac{1}{\alpha^3} - \frac{2}{\alpha^3} + \frac{2}{\alpha^3} \right] = \frac{1}{\alpha^2} \quad \text{Hence } \sigma = \frac{1}{\alpha}$$

Thus for the exponential distribution

$$\text{Mean } (\mu) = \frac{1}{\alpha} ; \quad \text{S.D } (\sigma) = \frac{1}{\alpha}$$

Normal Distribution.

The continuous probability distribution having the probability density funcⁿ $f(x)$ given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

Where $-\infty < x < \infty$, $-\infty < \mu < \infty$ & $\sigma > 0$ is known as the normal distribution.

Evidently $f(x) \geq 0$

$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-\mu)^2/2\sigma^2} dx$$

Putting $t = \frac{x-\mu}{\sqrt{2}\sigma}$ (or) $x = \mu + \sqrt{2}\sigma t$, We have $dx = \sqrt{2}\sigma dt$

t also varies from $-\infty$ to ∞

$$\text{Hence } \int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2} \sqrt{2}\sigma dt$$

$$\text{ie., } \int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sqrt{\pi}} \cdot 2 \int_0^{\infty} e^{-t^2} dt$$

But $\int_0^{\infty} e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$ by gamma funcⁿ.

$$\therefore \int_{-\infty}^{\infty} f(x) dx = \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1.$$

Thus $f(x)$ represents a probability density funcⁿ.

• Mean & Standard Deviation of a Normal distribution.

$$\text{Mean } (\mu) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x \cdot e^{-(x-\mu)^2/2\sigma^2} dx$$

Putting $t = \frac{x-\mu}{\sqrt{2}\sigma}$ (or) $x = \mu + \sqrt{2}\sigma t$, We have $dx = \sqrt{2}\sigma dt$

t also varies from $-\infty$ to ∞

$$\begin{aligned}\text{Mean} &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sqrt{2}\sigma t) e^{-t^2} \sqrt{2}\sigma dt \\ &= \frac{\mu}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt + \sigma \frac{\sqrt{2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} t e^{-t^2} dt\end{aligned}$$

$$\text{Mean} = \frac{2\mu}{\sqrt{\pi}} \int_0^{\infty} e^{-t^2} dt + \sigma \frac{\sqrt{2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} t e^{-t^2} dt$$

Here the second integral is zero by a standard property
Since $t e^{-t^2}$ is an odd funcⁿ.

$$\text{Hence, Mean} = \frac{2\mu}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} + 0 = \mu.$$

Thus mean = μ .

Hence We can say that the mean of the normal distribution is equal to the mean of the given distribution.

$$\begin{aligned}\text{Variance}(\sigma^2) &= \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x-\mu)^2 e^{-(x-\mu)^2/2\sigma^2} dx\end{aligned}$$

Substituting $t = \frac{x-\mu}{\sqrt{2}\sigma}$ We have as in the earlier case,

$$\begin{aligned}\text{Variance} &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} 2\sigma^2 t^2 e^{-t^2} \sqrt{2}\sigma dt \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2} dt \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \cdot 2 \int_0^{\infty} t^2 e^{-t^2} dt \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} t (2t e^{-t^2}) dt\end{aligned}$$

$$\text{WKT } \int u v dt = u \int v dt - \int \left(\int v dt \right) \cdot u' dt$$

Taking $u = t$, $v = 2t \bar{e}^{t^2}$ & noting that $\int v dt = -\bar{e}^{t^2}$ We now have,
 variance = $\frac{2\sigma^2}{\sqrt{\pi}} \left\{ \left[t (-\bar{e}^{t^2}) \right]_0^\infty - \int_0^\infty -(\bar{e}^{t^2}) \cdot 1 dt \right\}$

$$\text{Variance} = \frac{2\sigma^2}{\sqrt{\pi}} \left\{ 0 + \int_0^\infty \bar{e}^{t^2} dt \right\} = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \sigma^2$$

Thus variance = σ^2 .

Hence We can say that the variance/S.D of the normal distribution is equal to the variance/S.D of the given distribution.

Standard Normal Distribution.

$$\text{We have } P(a \leq x \leq b) = \int_a^b f(x) dx$$

In the case of normal distribution We have,

$$P(a \leq x \leq b) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b \bar{e}^{-(x-\mu)^2/2\sigma^2} dx \quad \text{--- (1)}$$

The integral in the RHS of this eqⁿ cannot be evaluated by known methods of integration & We have to employ the technique of numerical integration which becomes tedious. Hence We think of standardization & the same is as follows.

Putting $z = \frac{x-\mu}{\sigma}$ (or) $x = \mu + \sigma z$ We have $dx = \sigma dz$

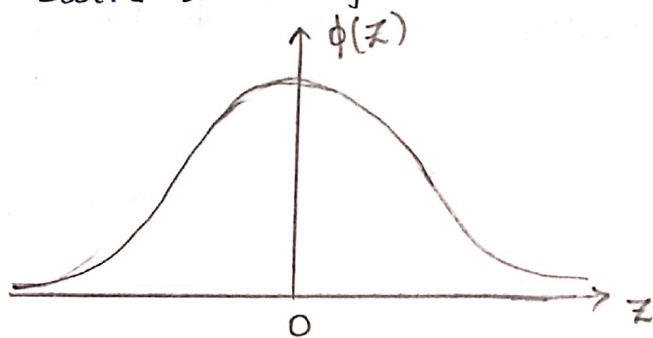
Let $z_1 = \frac{a-\mu}{\sigma}$ & $z_2 = \frac{b-\mu}{\sigma}$ be the values of z corresponding to $x=a$ & $x=b$. The integral in (1) assumes the following form.

$$P(a \leq x \leq b) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} \bar{e}^{-z^2/2} dz$$

$$\text{Hence, } P(a \leq x \leq b) = P(z_1 \leq z \leq z_2) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} \bar{e}^{-z^2/2} dz \quad \text{--- (2)}$$

If $F(z) = \frac{1}{\sqrt{2\pi}} \bar{e}^{-z^2/2}$ (standard normal probability density funcⁿ),

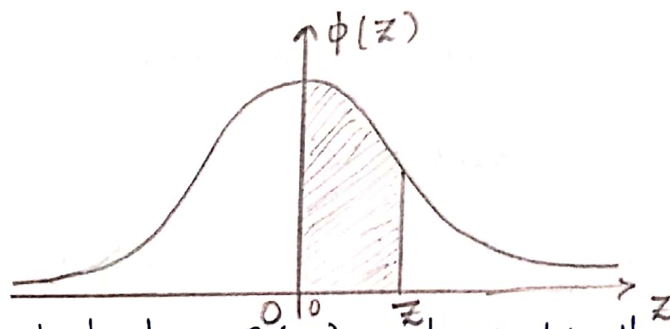
It may be observed that this is same as the p.d.f of the normal distribution with $\mu=0$ & $\sigma=1$. Thus we can say that the normal probability density funcⁿ with $\mu=0$ & $\sigma=1$ is the standard normal probability density funcⁿ. $Z = \frac{x-\mu}{\sigma}$ is called the standard normal variate (S.N.V) & $F(Z)$ is called the ^{std.} normal curve which is symmetrical about the line $Z=0$. The curve is as follows.



The integral in the RHS of (2) geometrically represents the area bounded by the standard normal curve $F(z)$ b/w $z=z_1$ & $z=z_2$. Further in particular if $z_1=0$ we have

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2} dz$$

This represents the area under the standard normal curve from $z=0$ to z .



$\phi(z)$ also denoted by $A(z)$ represents the area (shaded portion) as shown in the figure. Since the total area is 1, the area on either side of $z=0$ is 0.5.

Problems

(Exponential probability)

Find which of the following functions is a probability density funcⁿ.

$$(i) f_1(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(ii) f_2(x) = \begin{cases} 2x, & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(iii) f_3(x) = \begin{cases} |x|, & |x| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(iv) f_4(x) = \begin{cases} 2x, & 0 < x \leq 1 \\ 4-4x, & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Conditions for a P.d.f are $f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x) dx = 1$.

(i) Clearly $f(x) \geq 0$

$$\int_{-\infty}^{\infty} f_1(x) dx = \int_0^1 f_1(x) dx = \int_0^1 2x dx = [x^2]_0^1 = 1$$

$\therefore f_1(x)$ is a P.d.f.

(ii) The given function can be written in the form.

$$f_2(x) = \begin{cases} 2x, & -1 < x < 0 \\ 2x, & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

In $-1 < x < 0$, $f_2(x) = 2x$ is less than zero.

$$\text{Further } \int_{-\infty}^{\infty} f_2(x) dx = \int_{-1}^1 2x dx = [x^2]_{-1}^1 = 0$$

Both the conditions are not satisfied.

$\therefore f_2(x)$ is not a P.d.f.

(iii) Evidently $f_3(x) = |x| \geq 0$

$$\int_{-\infty}^{\infty} f_3(x) dx = \int_{-1}^1 f_3(x) dx = \int_{-1}^1 |x| dx$$

$$\text{But } |x| = \begin{cases} -x, & -1 < x < 0 \\ +x, & 0 < x < 1 \end{cases}$$

$$\begin{aligned}\therefore \int_{-\infty}^{\infty} f_3(x) dx &= \int_{-1}^0 -x dx + \int_0^1 x dx \\ &= -\left[\frac{x^2}{2}\right]_{-1}^0 + \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2} + \frac{1}{2} = 1\end{aligned}$$

$\therefore f_3(x)$ is a P.d.f

(iv) The given function

$$f_4(x) = 2x > 0 \text{ in } 0 < x \leq 1$$

But $f_4(x) = 4 - 4x$ is negative in $1 < x < 2$

The first condition is not satisfied.

$\therefore f_4(x)$ is not a P.d.f.

Q> Find the value of c s.t. $f(x) = \begin{cases} x/6 + c, & 0 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$ is a P.d.f

Also find $P(1 \leq x \leq 2)$

$f(x) \geq 0$ if $c \geq 0$; Also we must have $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\text{i.e., } \int_0^3 \left(\frac{x}{6} + c\right) dx = 1$$

$$\text{i.e., } \left[\frac{x^2}{12} + cx\right]_0^3 = 1$$

$$\text{i.e., } \frac{3}{4} + 3c = 1 \quad \therefore c = \frac{1}{12}$$

$$\begin{aligned}\text{Now } P(1 \leq x \leq 2) &= \int_1^2 f(x) dx \\ &= \int_1^2 \left(\frac{x}{6} + \frac{1}{12}\right) dx \\ &= \left[\frac{x^2}{12} + \frac{x}{12}\right]_1^2\end{aligned}$$

$$= \frac{1}{12} [(4+2) - (1+1)] = \frac{1}{3}$$

Thus $P(1 \leq x \leq 2) = \frac{1}{3}$

3) Find the constant k s.t. $f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$ is a p.d.f

Also compute (i) $P(1 < x < 2)$ (ii) $P(x \leq 1)$ (iii) $P(x > 1)$
(iv) Mean (v) Variance.

$f(x) \geq 0$ if $k \geq 0$. Also we must have, $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\text{i.e., } \int_0^3 kx^2 dx = 1$$

$$\text{i.e., } \left[\frac{kx^3}{3} \right]_0^3 = 1 \quad (\text{or}) \quad 9k = 1 \quad \therefore k = \frac{1}{9}$$

$$(i) \quad P(1 < x < 2) = \int_1^2 f(x) dx = \int_1^2 \frac{x^2}{9} dx = \left[\frac{x^3}{27} \right]_1^2 = \frac{7}{27}$$

$$(ii) \quad P(x \leq 1) = \int_0^1 f(x) dx = \int_0^1 \frac{x^2}{9} dx = \left[\frac{x^3}{27} \right]_0^1 = \frac{1}{27}$$

$$(iii) \quad P(x > 1) = \int_1^3 f(x) dx = \int_1^3 \frac{x^2}{9} dx = \left[\frac{x^3}{27} \right]_1^3 = \frac{26}{27}$$

$$(iv) \quad \text{Mean} = \mu = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_0^3 x \cdot \frac{x^2}{9} dx = \left[\frac{x^4}{36} \right]_0^3 = \frac{81}{36} = \frac{9}{4}$$

$$(v) \quad \text{Variance} = V = \int_{-\infty}^{\infty} x^2 f(x) dx - (\mu)^2$$

$$= \int_0^3 x^2 \cdot \frac{x^2}{9} dx - \left(\frac{9}{4}\right)^2$$

$$V = \left[\frac{x^5}{45} \right]_0^3 - \frac{81}{16} = \frac{81}{15} - \frac{81}{16} = \frac{81}{240} = \frac{27}{80}$$

4) Find k s.t. $f(x) = \begin{cases} kxe^{-x}, & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ is a p.d.f. Find the

mean.

$f(x) \geq 0$ if $k \geq 0$. Also we must have $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\text{i.e., } \int_0^1 kxe^{-x} dx = 1$$

Applying Bernoulli's rule we have,

$$k [x(-e^x) - (1)(e^x)]_0^1 = 1$$

$$\text{ie., } k \left[-\frac{1}{e} - \left(\frac{1}{e} - 1 \right) \right] = 1,$$

$$\text{ie., } k \left[1 - \frac{2}{e} \right] = 1$$

$$\therefore k = \frac{e}{e-2}$$

$$\text{Mean} = \mu = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_0^1 x \cdot \frac{e}{e-2} x e^x dx$$

$$= \frac{e}{e-2} \int_0^1 x^2 e^x dx$$

$$= \frac{e}{e-2} \left[x^2 (-e^x) - (2x)(e^x) + 2(-e^x) \right]_0^1$$

$$\mu = \frac{e}{e-2} \left[-\frac{1}{e} - \frac{2}{e} - 2 \left(\frac{1}{e} - 1 \right) \right]$$

$$\mu = \frac{e}{e-2} \left(2 - \frac{5}{e} \right) = \underline{\underline{\frac{2e-5}{e-2}}}$$

5) Is the following funcⁿ a density funcⁿ?

$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

If so, determine the probability that the variate having this density will fall in the interval (1, 2).

We observe $f(x) \geq 0$. Also we must have $\int_{-\infty}^{\infty} f(x) dx = 1$.

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx$$

$$= 0 + \int_0^{\infty} e^{-x} dx = [-e^{-x}]_0^{\infty} = -(0-1) = 1$$

Hence $f(x)$ is a probability density funcⁿ.

The probability that the variate having this density will fall in the interval $(1, 2)$ is to compute $P(1 < x < 2)$

$$\therefore P(1 < x < 2) = \int_1^2 f(x) dx = -[e^{-x}]_1^2 = -(e^{-2} - e^{-1})$$

$$\text{Thus } P(1 < x < 2) = \left(\frac{1}{e} - \frac{1}{e^2}\right) = 0.2325.$$

6) A random variable x has the following density funcⁿ

$$P(x) = \begin{cases} kx^2, & -3 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate k & find (i) $P(1 \leq x \leq 2)$ (ii) $P(x \leq 2)$ (iii) $P(x > 1)$

$P(x) \geq 0$ if $k \geq 0$ & also we must have $\int_{-\infty}^{\infty} P(x) dx = 1$

$$\text{i.e., } \int_{-3}^3 kx^2 dx = 1$$

$$\text{or } \left[\frac{kx^3}{3} \right]_{-3}^3 = 1 \quad \therefore k = \frac{1}{18}$$

$$(i) P(1 \leq x \leq 2) = \int_1^2 \frac{x^2}{18} dx = \left[\frac{x^3}{54} \right]_1^2 = \frac{1}{54} (8 - 1) = \frac{7}{54}$$

$$(ii) P(x \leq 2) = \int_{-3}^2 \frac{1}{18} x^2 dx = \frac{1}{18} \left[\frac{x^3}{3} \right]_{-3}^2 = \frac{1}{54} (8 + 27) = \frac{35}{54}$$

$$(iii) P(x > 1) = \int_1^3 \frac{1}{18} x^2 dx = \frac{1}{18} \left[\frac{x^3}{3} \right]_1^3 = \frac{1}{54} (27 - 1) = \frac{26}{54} = \frac{13}{27}$$

7) Find the c.d.f for the following p.d.f of a random variable x .

$$(i) f(x) = \begin{cases} 6x - 6x^2, & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad (ii) f(x) = \begin{cases} \frac{x}{4} e^{-x/2}, & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

(iii) Exponential distribution.

If $f(x)$ is the p.d.f then the c.d.f $= F(x) = \int_{-\infty}^x f(x) dx$

$$(i) F(x) = \int_{-\infty}^x f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= 0 + \int_0^x f(x) dx$$

$$\therefore F(x) = \int_0^x (6x - 6x^2) dx = [3x^2 - 2x^3]_0^x = 3x^2 - 2x^3$$

$$\therefore \text{c.d.f} = 3x^2 - 2x^3 \text{ if } 0 \leq x \leq 1.$$

$$(ii) F(x) = \int_0^x \frac{x}{4} e^{-x/2} dx$$

$$= \frac{1}{4} \left[x \cdot \frac{e^{-x/2}}{-1/2} - 1 \cdot \frac{e^{-x/2}}{1/4} \right]_0^x$$

$$= \frac{1}{4} [-2(xe^{-x/2} - 0) - 4(e^{-x/2} - 1)]$$

$$\therefore \text{c.d.f} = 1 - e^{-x/2} - (x/2) e^{-x/2}, \text{ if } 0 < x < \infty$$

$$(iii) f(x) = \begin{cases} \alpha e^{-\alpha x} & , 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

is the p.d.f of the

exponential distribution.

$$F(x) = \int_{-\infty}^x f(x) dx = \int_0^x \alpha e^{-\alpha x} dx = -[e^{-\alpha x}]_0^x = 1 - e^{-\alpha x}$$

$$\therefore \text{c.d.f} = 1 - e^{-\alpha x} \text{ if } 0 < x < \infty$$

8) A continuous random variable has the distribution funcⁿ.

$$F(x) = \begin{cases} 0 & , x \leq 1 \\ c(x-1)^4 & , 1 \leq x \leq 3 \\ 1 & , x > 3 \end{cases}$$

Find c & also the p.d.f

$$\text{W.K.T the p.d.f } f(x) = \frac{d}{dx} [F(x)]$$

$$\therefore f(x) = \begin{cases} 0 & , x \leq 1 \\ 4c(x-1)^3 & , 1 \leq x \leq 3 \\ 0 & , x > 3 \end{cases}$$

$$f(x) \geq 0 \text{ for } c \geq 0 \text{ \& \; we must have } \int_{-\infty}^{\infty} f(x) dx = 1$$

Hence we must have $\int_1^3 4c(x-1)^3 dx = 1$

$$\text{ie., } [c(x-1)^4]_1^3 = 1,$$

$$\text{ie., } 16c = 1 \quad \therefore c = 1/16$$

Thus the p.d.f $f(x) = (x-1)^3/4$ Where $1 \leq x \leq 3$.

9) Find k so that the following funcⁿ can serve as a Probability density funcⁿ of a random variable.

$$f(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ kx e^{-4x^2} & \text{for } x > 0. \end{cases}$$

Put $4x^2 = t$
 $8x dx = dt \quad t \rightarrow 0 \text{ to } \infty$
 $k = 8.$

10) A random variable x has the density funcⁿ $f(x) = \frac{k}{1+x^2}$, $-\infty < x < \infty$. Determine k & hence evaluate

(i) $P(x \geq 0)$ (ii) $P(0 < x < 1)$.

11) The kilometre run (in thousands of k.ms) without any sort of problem in respect of a certain vehicle is a random variable having p.d.f.

$$f(x) = \begin{cases} \frac{1}{40} e^{-x/40} & , x \geq 0 \\ 0 & , x \leq 0 \end{cases}$$

Find the probability that the vehicle is trouble free.

i) at least for 25000 k.ms ii) at most for 25000 k.ms

iii) b/w 16000 to 32000 k.ms.

Here x is the random variable representing kilometre in multiples of 1000 regarding trouble free run by the vehicle.

(i) To find $P(x \geq 25)$

$$P(x \geq 25) = 1 - P(x < 25)$$

$$= 1 - \int_0^{25} \frac{1}{40} e^{-x/40} dx$$

$$= 1 + [e^{-x/40}]_0^{25} = 1 + [e^{-5/8} - 1] = e^{-5/8}$$

$$\therefore P(x \geq 25) = 0.5353.$$

ii) To find $P(x \leq 25)$

$$\begin{aligned} P(x \leq 25) &= \int_0^{25} \frac{1}{40} e^{-x/40} dx \\ &= \left[-e^{-x/40} \right]_0^{25} = -e^{-5/8} + 1 \end{aligned}$$

$$\therefore P(x \leq 25) = 0.4647.$$

iii) To find $P(16 \leq x \leq 32)$

$$\begin{aligned} P(16 \leq x \leq 32) &= \int_{16}^{32} \frac{1}{40} e^{-x/40} dx \\ &= \left[-e^{-x/40} \right]_{16}^{32} = -e^{-4/5} + e^{-2/5} \end{aligned}$$

$$\therefore P(16 \leq x \leq 32) = 0.221.$$

12) If x is an exponential variate with mean 3 find

(i) $P(x > 1)$ (ii) $P(x < 3)$

The P.d.f of the exponential distribution is given by

$$f(x) = \begin{cases} \alpha e^{-\alpha x}, & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

The mean of this distribution is given by $1/\alpha$

By data, mean $= \frac{1}{\alpha} = 3 \quad \therefore \alpha = 1/3$

Hence $f(x) = \begin{cases} \frac{1}{3} e^{-x/3}, & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$

(i) $P(x > 1) = 1 - P(x \leq 1)$

$$= 1 - \int_0^1 f(x) dx$$

$$= 1 - \int_0^1 \frac{1}{3} e^{-x/3} dx = 1 + \left[e^{-x/3} \right]_0^1 = e^{-1/3}$$

$$\therefore P(x > 1) = e^{-1/3} = 0.7165$$

$$\begin{aligned}
 \text{(ii)} \quad P(x < 3) &= \int_0^3 f(x) dx \\
 &= \int_0^3 \frac{1}{3} e^{-x/3} dx = - \left[e^{-x/3} \right]_0^3 \\
 &= 1 - \frac{1}{e} = 0.6321
 \end{aligned}$$

$$\therefore P(x < 3) = \underline{0.6321}$$

13) If x is an exponential variate with mean 5, evaluate.

(i) $P(0 < x < 1)$ (ii) $P(-\infty < x < 10)$ (iii) $P(x \leq 0 \text{ or } x \geq 1)$.

$$\text{Mean} = \frac{1}{\alpha} = 5 \quad \therefore \alpha = \frac{1}{5}$$

$$\text{i)} \quad \int_0^1 f(x) dx = 1 - e^{-0.2} = 0.1813$$

$$\text{ii)} \quad \int_{-\infty}^0 f(x) dx + \int_0^{10} f(x) dx = 1 - \frac{1}{e^2} = 0.8647$$

$$\begin{aligned}
 \text{iii)} \quad P(x \leq 0) + P(x \geq 1) \\
 = 0 + \int_1^{\infty} f(x) dx = \int_1^{\infty} \frac{1}{5} e^{-x/5} dx = - (0 - e^{-0.2}) = 0.8187
 \end{aligned}$$

14) The sales per day in a shop is exponentially distributed with the average sale amounting to Rs. 100 & net profit is 8%. Find the probability that the net profit exceeds Rs. 30 on two consecutive days.

Let x be the random variable of the sale in the shop. Since x is an exponential variate the p.d.f

$$f(x) = \alpha e^{-\alpha x}, \quad x > 0$$

$$\text{Mean} = \frac{1}{\alpha} = 100 \quad \therefore \alpha = \frac{1}{100} = 0.01$$

$$\text{Hence } f(x) = 0.01 e^{-0.01x}, \quad x > 0.$$

Let A be the amount for which profit is 8%.

$$\Rightarrow A \cdot \frac{8}{100} = 30 \quad \therefore A = 375$$

Probability of profit exceeding Rs. 30 is equal to
 $1 - \text{Prob}(\text{Profit} \leq \text{Rs. } 30)$

$$= 1 - \text{Prob}(\text{sales} \leq \text{Rs. } 375)$$

$$= 1 - \int_0^{375} (0.01) e^{-0.01x} dx = 1 + [e^{-0.01x}]_0^{375} = e^{-3.75}$$

The probability that profit exceeds Rs. 30 on a single day is $e^{-3.75}$

Thus the probability that it repeats on the following day is $e^{-3.75} \cdot e^{-3.75} = e^{-7.5} = 0.00055$.

15) In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for:

(i) 10 minutes or more (ii) Less than 10 minutes

(iii) b/w 10 & 12 minutes.

The p.d.f of the exponential distribution is given by

$$f(x) = \alpha e^{-\alpha x}, \quad x > 0 \quad \& \quad \text{the mean} = \frac{1}{\alpha}$$

$$\text{By data } \frac{1}{\alpha} = 5 \quad \therefore \alpha = \frac{1}{5} \quad \& \quad \text{hence } f(x) = \frac{1}{5} e^{-x/5}$$

$$(i) \quad P(x \geq 10) = \int_{10}^{\infty} \frac{1}{5} \cdot e^{-x/5} dx = -[e^{-x/5}]_{10}^{\infty}$$

$$\therefore P(x \geq 10) = -(0 - e^{-2}) = e^{-2} = 0.1353$$

$$(ii) \quad P(x < 10) = \int_0^{10} \frac{1}{5} \cdot e^{-x/5} dx = -[e^{-x/5}]_0^{10}$$

$$\therefore P(x < 10) = -(e^{-2} - 1) = 1 - e^{-2} = 0.8647$$

$$(iii) \quad P(10 < x < 12) = \int_{10}^{12} \frac{1}{5} \cdot e^{-x/5} dx = -[e^{-x/5}]_{10}^{12}$$

$$\therefore P(10 < x < 12) = -(e^{-12/5} - e^{-2}) = 0.0446$$

Worked Problems on Normal Distribution.

We have to said that $\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2} dz$

represents the area under the standard normal curve from 0 to z . Tabulated values which gives the area for different +ve values of z is readily available. Such a table analogous to the format of a logarithmic table is called normal probability table. We present a few illustrations geometrically, theoretically & write the value by making use of the table.

$$(1) \int_{-\infty}^{\infty} \phi(z) dz = 1 \quad (2) \int_{-\infty}^0 \phi(z) dz = \int_0^{\infty} \phi(z) dz = \frac{1}{2}$$

These results in the equivalent form are as follows.

$$(1) P(-\infty \leq z \leq \infty) = 1 \quad (2) P(-\infty \leq z \leq 0) = \frac{1}{2}$$

$$(3) P(0 \leq z \leq \infty) \quad \text{or} \quad P(z \geq 0) = \frac{1}{2}$$

$$\text{Also } P(-\infty < z < z_1) = P(-\infty < z \leq 0) + P(0 \leq z < z_1)$$

$$\text{ie., } P(z < z_1) = 0.5 + \phi(z_1) \quad \text{--- (4)}$$

$$\text{Also } P(z > z_2) = P(z \geq 0) - P(0 \leq z < z_2)$$

$$\text{ie., } P(z > z_2) = 0.5 - \phi(z_2) \quad \text{--- (5)}$$

✓ Evaluate the following probabilities with the help of normal probability tables.

$$(i) P(z \geq 0.85)$$

$$(ii) P(-1.64 \leq z \leq -0.88)$$

$$(iii) P(z \leq -2.43)$$

$$(iv) P(|z| \leq 1.94)$$

$$(i) P(z \geq 0.85) = P(z \geq 0) - P(z \leq 0.85)$$

$$= 0.5 - \phi(0.85)$$

$$= 0.5 - 0.3023 = 0.1977$$

$$\therefore P(z \geq 0.85) = 0.1977$$

$$(ii) P(-1.64 \leq Z \leq -0.88)$$

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By symmetry, $P(-1.64 \leq Z \leq -0.88) = P(0.88 \leq Z \leq 1.64)$

$$\begin{aligned} \therefore P(-1.64 \leq Z \leq -0.88) &= P(0 \leq Z \leq 1.64) - P(0 \leq Z \leq 0.88) \\ &= \phi(1.64) - \phi(0.88) \end{aligned}$$

$$= 0.4495 - 0.3106 = 0.1389$$

$$\therefore P(-1.64 \leq Z \leq -0.88) = 0.1389$$

Remark: In this case, using the concept of symmetry we can directly write $\phi(1.64) - \phi(0.88)$

$$(iii) P(Z \leq -2.43)$$

$$P(Z \leq -2.43) = P(Z \geq 2.43)$$

$$= P(Z \geq 0) - P(Z \leq 2.43)$$

$$= 0.5 - \phi(2.43)$$

$$= 0.5 - \phi(2.43)$$

$$= 0.5 - 0.4925 = 0.0075$$

$$\therefore P(Z \leq -2.43) = 0.0075$$

$$(iv) P(|Z| \leq 1.94) = P(-1.94 \leq Z \leq 1.94)$$

$$= 2P(0 \leq Z \leq 1.94)$$

$$= 2\phi(1.94) = 2(0.4738) = 0.9476$$

$$\therefore P(|Z| \leq 1.94) = \underline{0.9476}$$

2) If X is a normal variate with mean 30 & SD 5 find the probability that

$$(i) 26 \leq X \leq 40$$

$$(ii) X \geq 45$$

We have standard normal variate (S.N.V) $Z = \frac{X - \mu}{\sigma} = \frac{X - 30}{5}$

$$(i) \text{ To find } P(26 \leq X \leq 40)$$

$$\text{If } X = 26, Z = -0.8; \text{ If } X = 40, Z = 2$$

Hence we need to find $P(-0.8 \leq Z \leq 2)$

$$\begin{aligned}
 P(-0.8 \leq z \leq 2) &= P(-0.8 \leq z \leq 0) + P(0 \leq z \leq 2) \\
 &= P(0 \leq z \leq 0.8) + P(0 \leq z \leq 2) \\
 &= \phi(0.8) + \phi(2) \\
 &= 0.2881 + 0.4772 = 0.7653
 \end{aligned}$$

$$\therefore P(26 \leq x \leq 40) = 0.7653$$

(ii) To find $P(x \geq 45)$

If $x = 45$, $z = 3$ & hence we have to find $P(z \geq 3)$

$$\begin{aligned}
 P(z \geq 3) &= P(z \geq 0) - P(z \leq 3) \\
 &= 0.5 - \phi(3) \\
 &= 0.5 - 0.4987 = 0.0013
 \end{aligned}$$

$$\therefore P(x \geq 45) = 0.0013.$$

3) If x is normally distributed with mean 12 & S.D 4, find the following.

(i) $P(x \geq 20)$ (ii) $P(x \leq 20)$

$$\text{We have S.N.V } z = \frac{x - \mu}{\sigma} = \frac{x - 12}{4}$$

$$\text{If } x = 20, z = 2$$

We have to find $P(z \geq 2)$ & $P(z \leq 2)$

$$\begin{aligned}
 \text{Now } P(x \geq 20) &= P(z \geq 2) - P(z \leq 2) \\
 &= 0.5 - \phi(2) \\
 &= 0.5 - 0.4772 = 0.0228
 \end{aligned}$$

$$\begin{aligned}
 \text{Also } P(z \leq 2) &= P(-\infty \leq z \leq 0) + P(0 \leq z \leq 2) \\
 &= 0.5 + \phi(2) \\
 &= 0.5 + 0.4772 = 0.9772
 \end{aligned}$$

Thus $P(x \geq 20) = 0.0228$ & $P(x \leq 20) = 0.9772$.

4) The marks of 1000 students in an examination follows a normal distribution with mean 70 & standard deviation 5. Find the no of students whose marks will be
(i) less than 65 (ii) more than 75 (iii) b/w 65 & 75.

Let x represents the marks of students.

By data $\mu = 70$, $\sigma = 5$ Hence s. n. v $z = \frac{x - \mu}{\sigma} = \frac{x - 70}{5}$

(i) If $x = 65$, $z = -1$ & We have to find $P(z < -1)$

$$P(z < -1) = P(z > 1)$$

$$= P(z \geq 0) - P(0 < z < 1)$$

$$= 0.5 - \phi(1) = 0.5 - 0.3413 = 0.1587.$$

$\therefore$ no of students scoring less than 65 marks

$$= 1000 \times 0.1587 = 158.7 \approx 159.$$

(ii) If $x = 75$, $z = 1$ & We have to find $P(z > 1)$

$$P(z > 1) = P(z \geq 0) - P(0 < z < 1)$$

$$= 0.5 - \phi(1) = 0.5 - 0.3413 = 0.1587.$$

$\therefore$ no of students scoring more than 75 marks

$$= 1000 \times 0.1587 = 158.7 \approx 159$$

(iii) We have to find $P(-1 < z < 1)$

$$P(-1 < z < 1) = 2P(0 < z < 1)$$

$$= 2\phi(1) = 2(0.3413) = 0.6826$$

$\therefore$ no of students scoring marks b/w 65 & 75

$$= 1000 \times 0.6826 = 682.6 \approx \underline{683}$$

- 5) In a test on electric bulbs, it was found that the life time of a particular brand was distributed normally with an average life of 2000 hours & S.D of 60 hours. If a firm purchases 2500 bulbs find the no of bulbs that are likely to last for (i) more than 2100 hours
(ii) Less than 1950 hours (iii) b/w 1900 to 2100 hours.

By data $\mu = 2000$, $\sigma = 60$

We have S.N.V $Z = \frac{x - \mu}{\sigma} = \frac{x - 2000}{60}$

(i) To find $P(x > 2100)$

If $x = 2100$, $Z = \frac{100}{60} \approx 1.67$

$$\begin{aligned} P(x > 2100) &= P(Z > 1.67) \\ &= P(Z \geq 0) - P(0 < Z < 1.67) \\ &= 0.5 - \phi(1.67) \\ &= 0.5 - 0.4525 = 0.0475 \end{aligned}$$

$\therefore$ no of bulbs that are likely to last for more than 2100 hours is $2500 \times 0.0475 = 118.75 \approx 119$.

(ii) To find $P(x < 1950)$

If $x = 1950$, $Z = \frac{-50}{60} = -0.83$

$$\begin{aligned} P(x < 1950) &= P(Z < -0.83) \\ &= P(Z > 0.83) \\ &= P(Z \geq 0) - P(0 < Z < 0.83) \\ &= 0.5 - \phi(0.83) \\ &= 0.5 - 0.2967 = 0.2033 \end{aligned}$$

$\therefore$ no of bulbs that are likely to last for less than 1950 hours is $2500 \times 0.2033 = 508.25 \approx 508$

(iii) To find $P(1900 < x < 2100)$

If $x = 1900$, $Z = -1.67$ & if $x = 2100$, $Z = 1.67$

$$P(1900 < x < 2100) = P(-1.67 < z < 1.67)$$

$$= 2P(0 < z < 1.67)$$

$$= 2\phi(1.67) = 2 \times 0.4525 = 0.905$$

$\therefore$ no. of bulbs that are likely to last b/w 1900 & 2100 hours

$$= 2500 \times 0.905 = 2262.5 \approx 2263.$$

6) In a normal distribution 31% of the items are under 45 & 8% of the items are over 64. Find the mean & S.D of the distribution.

Let μ & σ be the mean & S.D of the normal distribution

By data $P(x < 45) = 0.31$ & $P(x > 64) = 0.08$

We have S.N.V $z = \frac{x - \mu}{\sigma}$

When $x = 45$, $z = \frac{45 - \mu}{\sigma} = z_1$ (say)

$x = 64$, $z = \frac{64 - \mu}{\sigma} = z_2$ (say)

Thus we have,

$$P(z < z_1) = 0.31 \quad \& \quad P(z > z_2) = 0.08$$

ie., $0.5 + \phi(z_1) = 0.31$ & $0.5 - \phi(z_2) = 0.08$

$$\Rightarrow \phi(z_1) = -0.19 \quad \& \quad \phi(z_2) = 0.42$$

$$0.1915 (\approx 0.19) = \phi(0.5) \quad \& \quad 0.4192 (\approx 0.42) = \phi(1.4)$$

$$\therefore \phi(z_1) = -\phi(0.5) \quad \& \quad \phi(z_2) = \phi(1.4)$$

$$\Rightarrow z_1 = -0.5 \quad \& \quad z_2 = 1.4$$

ie., $\frac{45 - \mu}{\sigma} = -0.5$ & $\frac{64 - \mu}{\sigma} = 1.4$

(or) $\mu - 0.5\sigma = 45$ & $\mu + 1.4\sigma = 64$

By solving we get $\mu = 50$, $\sigma = 10$

Thus mean = 50 & SD = 10

7) In an examination 7% of students score less than 35% marks & 89% of students score less than 60% marks. Find the mean & S.D if the marks are normally distributed. It is given that if

$$P(Z) = \frac{1}{\sqrt{2\pi}} \int_0^Z e^{-z^2/2} dz \quad \text{then } P(1.2263) = 0.39 \quad \& \quad P(1.4757) = 0.43$$

Let μ & σ be the mean & S.D of the normal distribution.

By data we have $P(X < 35) = 0.07$, $P(X < 60) = 0.89$

We have S.N.V $Z = \frac{X - \mu}{\sigma}$

When $X = 35$, $Z = \frac{35 - \mu}{\sigma} = Z_1$ (say)

$X = 60$, $Z = \frac{60 - \mu}{\sigma} = Z_2$ (say)

Hence we have

$$P(Z < Z_1) = 0.07 \quad \& \quad P(Z < Z_2) = 0.89$$

ie., $0.5 + \phi(Z_1) = 0.07 \quad \& \quad 0.5 + \phi(Z_2) = 0.89$

$\therefore \phi(Z_1) = -0.43 \quad \& \quad \phi(Z_2) = 0.39$

Using the given data in the RHS of these we have,

$$\phi(Z_1) = -\phi(1.4757) \quad \& \quad \phi(Z_2) = \phi(1.2263)$$

$$\Rightarrow Z_1 = -1.4757 \quad \& \quad Z_2 = 1.2263$$

ie., $\frac{35 - \mu}{\sigma} = -1.4757 \quad \& \quad \frac{60 - \mu}{\sigma} = 1.2263$

(or) $\mu - 1.4757\sigma = 35 \quad \& \quad \mu + 1.2263\sigma = 60$

By solving we get $\mu = 48.65 \quad \& \quad \sigma = 9.25$

Thus mean = 48.65 & S.D = 9.25

8) Obtain the eqⁿ of the normal probability curve that may be fitted to the following data.

Variable	6	7	8	9	10	11	12
Frequency	3	6	9	13	8	5	4

The eqⁿ of the best fitting normal probability curve is

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

We need to compute μ & σ for the given frequency distribution.

$$\mu = \frac{\sum fx}{\sum f} = \frac{18 + 42 + 72 + 117 + 80 + 55 + 48}{48} = 9$$

$$\begin{aligned} \sigma^2 &= \frac{\sum fx^2}{\sum f} - (\mu)^2 \\ &= \frac{3(36) + 6(49) + 9(64) + 13(81) + 8(100) + 5(121) + 4(144)}{48} - 9^2 \end{aligned}$$

$$\sigma^2 = 2.5833 \quad \therefore \sigma = 1.607$$

$$\therefore f(x) = \frac{1}{(1.607)\sqrt{2\pi}} e^{-\frac{(x-9)^2}{5.1666}}$$

Thus the required eqⁿ of the normal probability curve is

$$f(x) = (0.24825) e^{-\frac{(x-9)^2}{5.1666}}$$